

Temperature-Based Correction of Triaxial Acceleration Error

Hyperparameter Tuning & Model Evaluation

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Abstract

This report evaluates whether a long short-term memory (LSTM) neural network can use thermal measurements to predict triaxial acceleration error in a sensor assembly. The analysis was based on 4,440,384 one-second observations divided into uninterrupted recording periods. Fifty-three temperature-derived predictors were constructed from direct measurements, first differences, spatial summaries, sensor-to-electronics differences, and temporal lags. The final model used a 40-second input sequence and three-fold chunk-based walk-forward validation, with dropout, L2 regularization, early stopping, and fold-specific scaling to reduce overfitting and data leakage. Validation performance was poor (mean $R^2 = -8.576$), and performance on held-out recording chunks also remained below a mean-only baseline ($R^2 = -1.032$; MAE = 0.000503 m/s²). These findings do not support the prediction that the current temperature-only LSTM provides a reliable correction model. The results instead indicate substantial regime variation, weaknesses in the final validation direction, and possible contamination from sequences spanning concatenated recording gaps. A revised evaluation design, stronger baseline models, axis-specific diagnostics, and broader hyperparameter tuning are required before substantive conclusions can be drawn.

Keywords: *LSTM; hyperparameter tuning; time-series validation; sensor calibration; temperature compensation; acceleration error*

1. Introduction

1.1 Background

Temperature-dependent drift can compromise the accuracy of inertial and acceleration measurements when sensor components experience changing or spatially uneven thermal conditions. Because thermal effects may accumulate gradually and differ across the sensor assembly, a useful correction model must capture both the instantaneous temperature state and its temporal evolution. This study evaluates whether a recurrent neural network can learn that relationship from multichannel temperature observations and generalize it to recording periods not used for estimation.

1.2 Research question, hypothesis, and prediction

Research question: Can temperature measurements from multiple locations on the sensor assembly be used to predict and correct temperature-related error in the x-, y-, and z-axis acceleration signals over time?

Hypothesis: Temperature and spatial temperature gradients systematically affect measured acceleration error; therefore, a sequence model using present, lagged, and rate-of-change temperature features should predict the three-axis error better than a mean-only baseline on unseen recording periods.

Prediction: If the hypothesis is supported, the tuned model should achieve positive held-out R^2 , low axis-specific MAE/RMSE, and stable performance across chronological folds and independent recording chunks. Predictions should also follow the timing and amplitude of observed changes rather than collapsing toward a central value.

2. Methodology

2.1 Data preparation and analytical design

The analysis moved from a single uninterrupted series to a recurrent neural network evaluated with time-aware splits and then with separate recording chunks. The notebook loads 4,440,384 one-second observations, identifies uninterrupted runs, and constructs a three-element target vector in x-y-z order. The final chunk-based experiment uses six later chunks for training (3,486,300 complete rows after feature construction) and five earlier chunks for testing (953,424 complete rows).

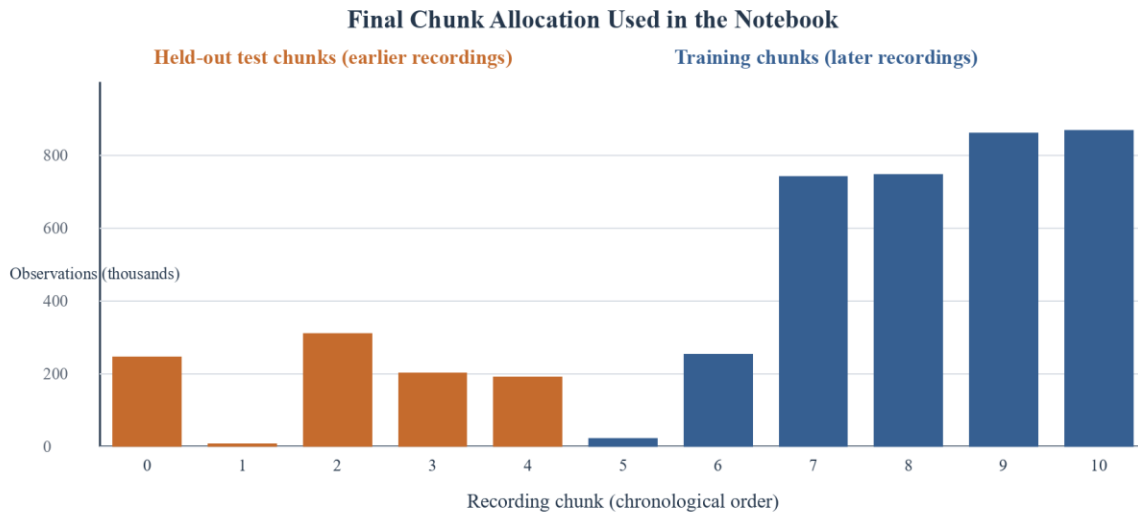


Figure 1. Chronological recording chunks used in the final analysis. Orange denotes earlier held-out chunks; blue denotes later training chunks.

2.2 Model specification: LSTM regression

A long short-term memory (LSTM) network was selected because both thermal conditions and sensor response are temporally dependent. Each input sample is a sequence of 53 temperature-derived features. The network contains 64 LSTM units, a 32-unit dense layer, and a three-unit linear output for simultaneous prediction of x, y, and z acceleration error. ReLU activation was used in the LSTM and dense hidden layer, with Adam optimization and mean squared error loss.

- Six direct temperature channels: T z, T x, T y, x sensor face, y sensor face, and top.
- Six first differences; temperature mean, standard deviation, and range; two sensor/electronics temperature differences; and 36 temperature lags at 1, 2, 5, 10, 30, and 60 seconds.
- Acceleration lags were intentionally excluded because the intended deployment may not have a stable or usable prior acceleration signal; this also reduces direct target leakage risk.

2.3 Hyperparameter tuning and overfitting controls

The notebook initially defined a grid of 2-6 walk-forward folds and sequence lengths of 10-60 seconds. A broad uninterrupted-series experiment executed the 30 combinations, while the final chunk-based experiment retained one selected configuration: three folds and a 40-second sequence. Other architectural values were held fixed rather than tuned.

Component	Value / range	Reason
Sequence length	10, 20, 30, 40, 50, 60 s; final 40 s	Tests the duration of thermal memory.
Walk-forward folds	2-6; final 3	Preserves order while checking temporal generalization.

Component	Value / range	Reason
Regularization	Dropout 0.30; recurrent dropout 0.20; L2 = 1e-4	Constrains an expressive network and limits co-adaptation.
Training	Up to 10 epochs; batch 256; early stopping patience 3	Stops when validation loss no longer improves.
Scaling	MinMaxScaler for X; StandardScaler for y	Fit inside each fold on training observations only.

3. Results

3.1 Model fit and held-out evaluation

Evaluation	R ²	MAE (m/s ²)	MSE ((m/s ²) ²)
Chunk CV fold 1	-12.544	0.000131	7.0 x 10 ⁻⁸
Chunk CV fold 2	-4.022	0.000192	1.0 x 10 ⁻⁷
Chunk CV fold 3	-9.160	0.000263	1.4 x 10 ⁻⁷
CV average	-8.576	0.000195	1.04 x 10 ⁻⁷
Held-out chunks	-1.032	0.000503	9.25 x 10 ⁻⁷

Interpretation. Every R² value is negative. The model therefore explains less held-out variance than predicting the relevant target mean, despite MAE values that look numerically small because the acceleration errors themselves are on the order of 10⁻⁴ to 10⁻³ m/s². Held-out MAE is about 2.6 times the cross-validation average, and held-out MSE is about 8.9 times the cross-validation average. This is evidence of weak generalization and distribution mismatch, not adequate model fit.

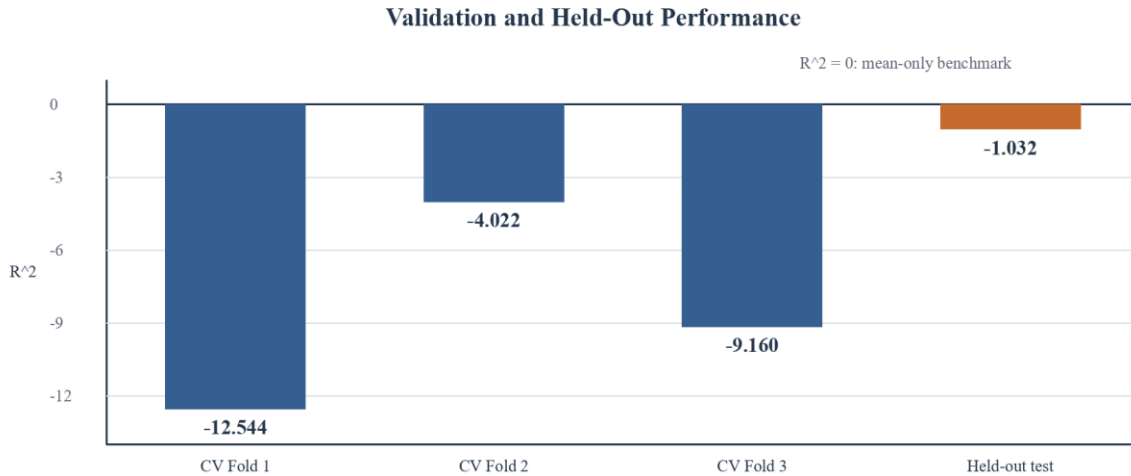


Figure 2. R^2 for the three chunk-based validation folds and the held-out test set. All values fall below the mean-only benchmark of zero.

3.2 Axis-specific behavior and diagnostic evidence

On the held-out chunks, signed mean percentage error was 22.60% for x, -64.69% for y, and 78.04% for z. Their simple average was 11.98%, but that aggregate is misleading because large positive and negative axis biases cancel. In addition, 12.33% of the 2,860,052 axis-level percentage errors were classified as boxplot outliers using the 1.5xIQR rule. Percentage error is unstable when true acceleration error approaches zero, so axis-level MAE/RMSE and R^2 should remain the primary metrics.

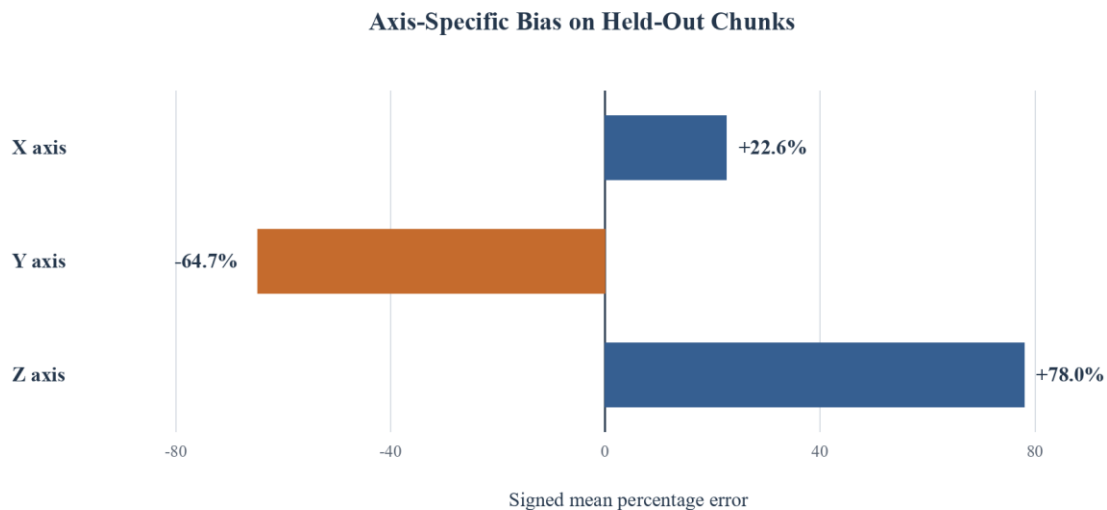


Figure 3. Signed mean percentage error by acceleration axis on the held-out chunks. Positive and negative axis biases partially cancel in the overall average.

3.3 Assumptions and validity checks

- Temporal dependence: an LSTM and chronological validation are appropriate in principle, but ordinary residual independence is not expected. Residual autocorrelation should be quantified with ACF plots or Ljung-Box tests.
- Stationarity / regime stability: the wide fold-to-fold performance and negative R^2 indicate that the temperature-error relationship changes across recording regimes. Residual-versus-time and residual-versus-temperature plots are still needed by axis.
- Data leakage: scalers are fit inside each training fold, which is good. However, the final code concatenates multiple recording chunks before forming sequences, so some 40-step windows can cross real time gaps. Sequences must be created within each chunk before concatenation.
- Validation direction: the final held-out chunks precede the training chunks. They are independent, but this is backward validation rather than prospective forecasting. A deployment claim requires training on earlier chunks and testing on later chunks.
- Model selection: the retained model is the best single validation fold, while the final chunk-based section evaluates only one configuration despite labels referring to nine models. Selection should use mean performance across all folds, followed by one refit on the complete training period.

3.4 Relationship to the research question and hypothesis

The current results do not support the prediction that the temperature-only LSTM generalizes well enough to correct acceleration error. They do not prove that temperature has no relationship with sensor error; instead, they show that the present feature set, validation design, and model configuration have not isolated a stable relationship across recording periods. The hypothesis remains plausible but is not supported by this fitted model.

4. Discussion

The analysis demonstrates that procedural controls against overfitting are necessary but not sufficient. Dropout, recurrent dropout, L2 penalties, early stopping, and training-only scaling were all incorporated, yet performance remained substantially below the baseline implied by $R^2 = 0$. The deterioration from cross-validation MAE to held-out MAE further indicates that the learned relationship did not transfer consistently across recording regimes. Accordingly, the current model should not be used to make operational corrections or substantive claims about the magnitude of temperature-induced acceleration error.

The negative results materially change the analysis plan. The immediate objective is no longer to select the best-performing LSTM configuration; it is to establish a leakage-resistant evaluation framework and determine whether any model can recover a stable, generalizable signal. In particular, sequences must respect recording boundaries, validation must proceed from earlier to later periods, and model selection must be based on aggregate performance across complete folds rather than the best single fold.

4.1 Further tuning and model evaluation

- Rebuild sequences separately within each uninterrupted chunk, then concatenate only completed sequences. Keep a final later-in-time chunk fully untouched until model selection is complete.
- Report axis-specific R^2 , MAE, RMSE, median absolute error, and residual plots. Replace signed mean percentage error with symmetric MAPE or scaled error only if a near-zero-safe business metric is needed.
- Fit baseline models: per-axis mean/last-value predictors, ridge or elastic-net regression, random forest, and gradient-boosted trees. These provide interpretable benchmarks and can reveal nonlinear temperature effects without recurrent complexity.
- For the LSTM, tune sequence length, units (e.g., 16/32/64), number of layers, learning rate, dropout, L2 penalty, and batch size using nested or blocked walk-forward validation. Choose by mean fold score, not the best fold.
- Test feature robustness by adding the currently excluded temperature channels one group at a time, and assess whether differencing or detrending the target improves regime stability. Any acceleration-lag experiment should be separated as a distinct deployment scenario.
- Check residual autocorrelation, heteroscedasticity across temperature ranges, and performance by recording chunk. If performance remains unstable, consider a state-space model, per-regime calibration, domain adaptation, or a physics-informed correction model.

5. Conclusion

The tuned LSTM did not provide an adequate model of temperature-related triaxial acceleration error. Its negative validation and test R^2 values show that it performed worse than a mean-only benchmark on unseen data, while substantial axis-specific bias and error dispersion further weaken the apparent 11.98% aggregate signed percentage error. Consequently, the stated prediction is not supported by the current evidence. The broader hypothesis that temperature contributes to acceleration error remains scientifically plausible, but it requires evaluation with a corrected time-series design, transparent baseline comparisons, more comprehensive tuning, and residual diagnostics. Until those steps are completed, the model should be regarded as exploratory rather than suitable for inference or deployment.

Appendix A. Data Dictionary and Codebook

Variable	Role	Description / units
Date	Time index	Original timestamp; converted to elapsed seconds within a recording chunk.

Variable	Role	Description / units
x [m/s ²]	Target	X-axis acceleration error, m/s ² .
y [m/s ²]	Target	Y-axis acceleration error, m/s ² .
z [m/s ²]	Target	Z-axis acceleration error, m/s ² .
T x [°C]	Predictor	Temperature associated with the x axis, °C.
T y [°C]	Predictor	Temperature associated with the y axis, °C.
T z [°C]	Predictor	Temperature associated with the z axis, °C.
bottom x1 side [°C]	Candidate predictor	Temperature at bottom x1 side, °C; excluded from final direct feature list.
x sensor face [°C]	Predictor	Temperature at x sensor face, °C.
y sensor face [°C]	Predictor	Temperature at y sensor face, °C.
bottom x2 side [°C]	Candidate predictor	Temperature at bottom x2 side, °C; excluded from final direct feature list.
x elect face [°C]	Derived-input source	Electronics-face temperature for x sensor difference, °C.
y ACQ face [°C]	Derived-input source	Acquisition-face temperature for y sensor difference, °C.
top [°C]	Predictor	Temperature at the top of the assembly, °C.
inside air [°C]	Candidate predictor	Internal air temperature, °C; excluded from final direct feature list.
external [°C]	Candidate predictor	External temperature, °C; excluded from final direct feature list.
Altitude [m]	Excluded	Altitude in meters; dropped before modeling.

Engineered features

Feature group	Count	Definition
Direct temperatures	6	T x, T y, T z, x sensor face, y sensor face, and top.
First differences	6	One-step change (dT) for each direct temperature.
Spatial summaries	3	Row-wise mean, standard deviation, and range of the six direct temperatures.
Temperature differences	2	x sensor face - x elect face; y sensor face - y ACQ face.
Lagged temperatures	36	Six direct temperatures at lags 1, 2, 5, 10, 30, and 60 seconds.
Total	53	Model input features at each time step.